A DISJOINT UNION OF TWO AFFINE SCHEMES IS AFFINE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 1252-1268, following the statement at 1245-1250.

Editorial note. The sentence 'This of course means that Spec(R) is the coproduct in the category of locally ringed spaces as well' is a universal-property reading of the displayed topological identification.

Allowed standard prerequisites (editorial).

- Canonical morphism into the spectrum of global sections. lemma-morphism-into-affine (same chapter, cited at ~1255). Builds X -> Spec(R) for R = Gamma(X, O_X).
- Spectrum of a product of rings. Algebra, Lemma algebra-lemma-spec-product (cross-chapter, cited at ~1257). Spec(O_X(U)) coprod Spec(O_X(V)) = Spec(R) as topological spaces.

Source reference. Original reference 00ED: focused-seeds/geometry/S9-schemes-disjoint-union-affines/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01I1: sources/stacks-schemes/pinned-original-source.tex, source lines 1058–1058. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a locally ringed space. Assume $X = U \amalg V$ with U and V open and such that U, V are affine schemes. Then X is an affine scheme.*

Proof. Set $R = \Gamma(X, \mathcal{O}_X)$. Note that $R = \mathcal{O}_X(U) \times \mathcal{O}_X(V)$ by the sheaf property. By Lemma 01I1 there is a canonical morphism of locally ringed spaces $X \to \mathrm{Spec}(R)$. By Algebra, Lemma 00ED we see that as a topological space $\mathrm{Spec}(\mathcal{O}_X(U)) \amalg \mathrm{Spec}(\mathcal{O}_X(V)) = \mathrm{Spec}(R)$ with the maps coming from the ring homomorphisms $R \to \mathcal{O}_X(U)$ and $R \to \mathcal{O}_X(V)$. This of course means that $\mathrm{Spec}(R)$ is the coproduct in the category of locally ringed spaces as well. By assumption the morphism $X \to \mathrm{Spec}(R)$ induces an isomorphism of $\mathrm{Spec}(\mathcal{O}_X(U))$ with U and similarly for V. Hence $X \to \mathrm{Spec}(R)$ is an isomorphism. □

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